External Connection Configuration

Unity Catalog · Azure Data Lake Storage Gen2 · Access Connector

Overview

Goal: Configure a Unity Catalog connection to external storage (Azure Data Lake Storage Gen2) using an Access Connector and External Location.

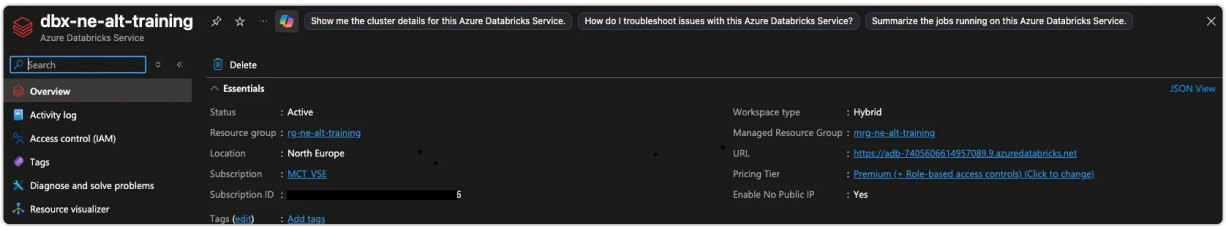
Prerequisites:

- Azure Databricks Workspace with Unity Catalog enabled
- Azure Data Lake Storage Gen2 (ADLS Gen2)
- Administrator permissions in both Azure Portal and Databricks Workspace

Step	Description
Part 1	Retrieve the Resource ID of the Access Connector
Part 2	Assign RBAC permissions on the Storage Account
Part 3	Create a Storage Credential in Databricks
Part 4	Create an External Location
Part 5	Configure access permissions

Part 1: Retrieve the Access Connector Resource ID

Navigate to **Azure Portal** → your Workspace → **Managed Resource Group**.



Azure Portal — Managed Resource Group view

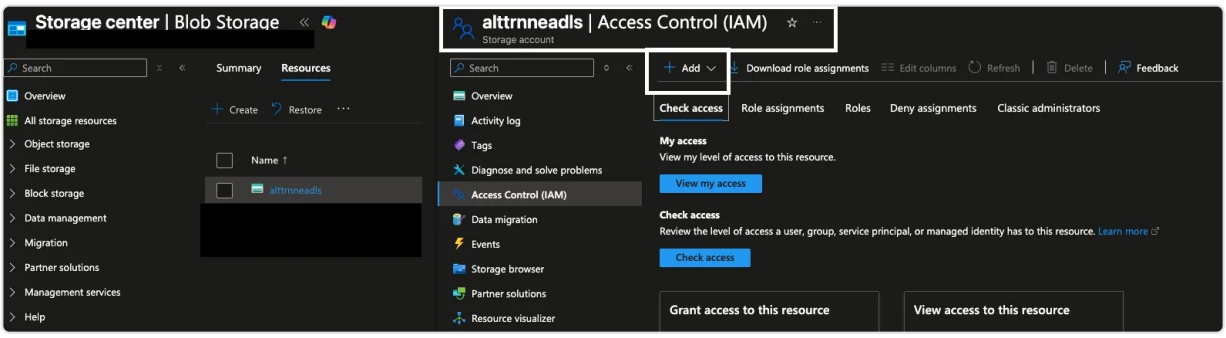
Inside the Managed Resource Group, locate the resource named **Access Connector for Azure Databricks** (e.g., `unity-access-connector`). Copy its **Resource ID** — you will need it in the following steps.

Access Connector — copy the Resource ID from the Properties blade

> **Note:** The Resource ID has the following format:
> `/subscriptions/<sub-id>/resourceGroups/<rg>/providers/Microsoft.Databricks/accessConnectors/<name>`

Part 2: Assign RBAC Permissions on the Storage Account

Navigate to your **Azure Data Lake Storage Gen2** account → **Access Control (IAM)**.

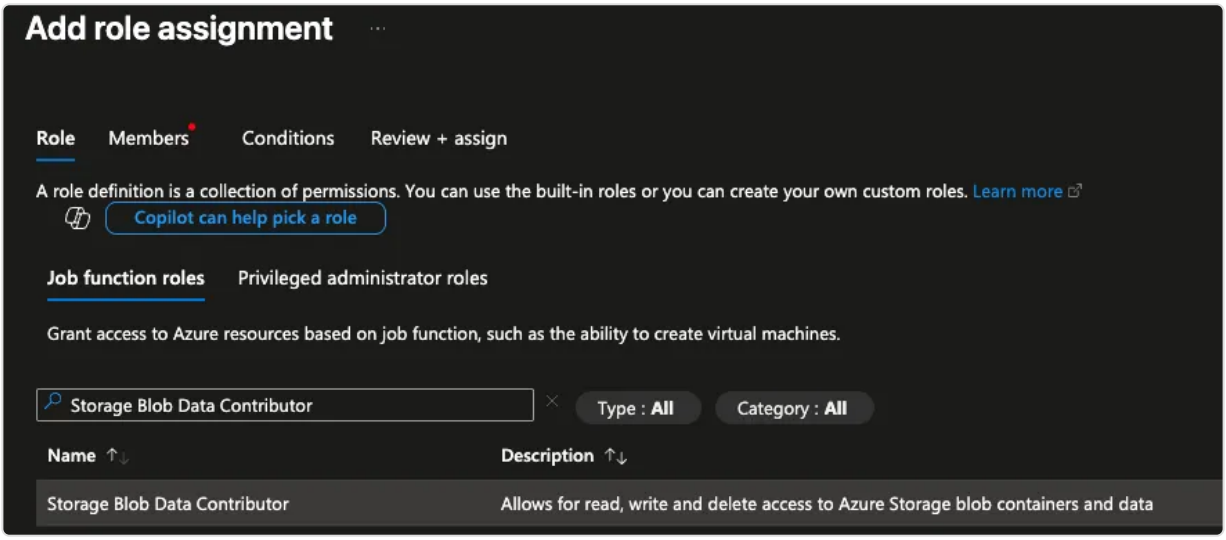


Storage Account — Access Control (IAM)

Add a role assignment following the **Principle of Least Privilege**:

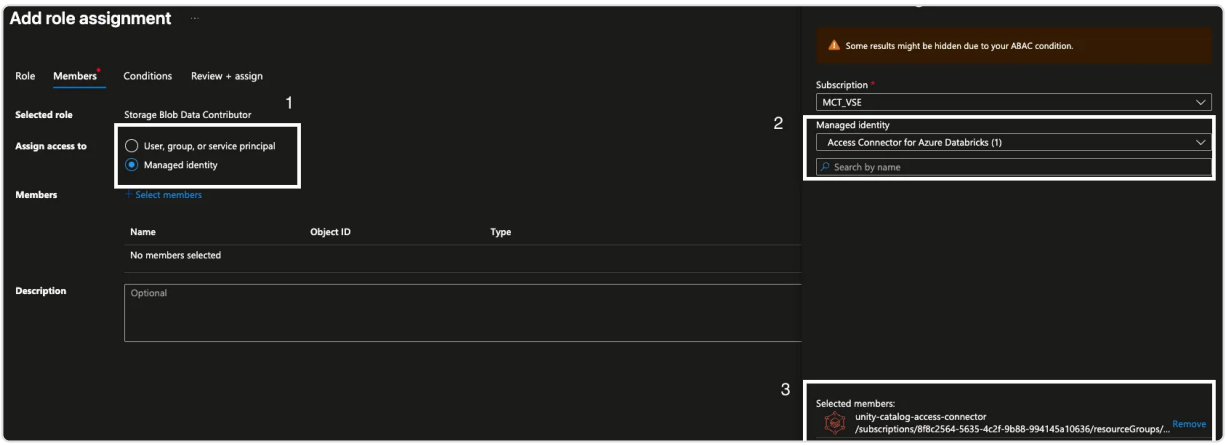
Role	Permissions	Use case
Storage Blob Data Reader	Read only	Production (read-only)
Storage Blob Data Contributor	Read + Write	Training / development

> ⚠️ **For this training** we assign **Storage Blob Data Contributor**. In a production environment, always apply the minimum required scope of permissions.



Role Assignment — select Storage Blob Data Contributor

Go to the **Members** tab → **Select members** → choose identity type **Managed Identity** → filter by **Access Connector for Azure Databricks** and select your connector.

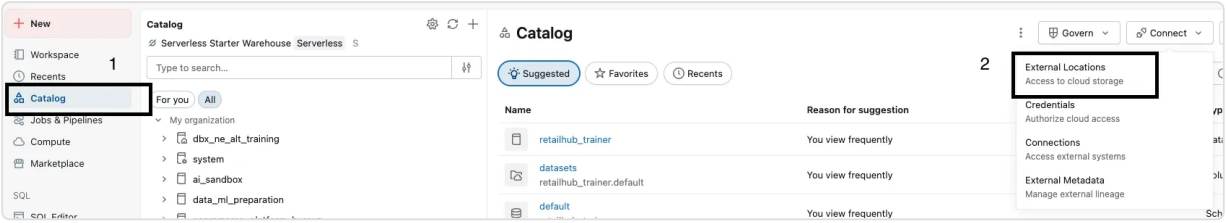


Members tab — select Managed Identity (Access Connector)

Click **Review + Assign** — done. The Access Connector now has permissions on the Storage Account.

Part 3: Create a Storage Credential in Databricks

Navigate to **Databricks Workspace** → **Catalog** → **External Data** → **Credentials**.



Databricks Catalog — External Data → Credentials

Create a new credential:

Field	Value
Credential name	Any name, e.g. `adls_training_credential`
Access connector ID	Resource ID copied in Part 1

> ✅ **Tip:** A Storage Credential is shared — it can be reused across multiple External Locations.

Part 4: Create an External Location

Return to the **External Locations** tab → click **Create External Location**.

Fill in the form:

Field	Value	Example
External location name	Custom name	`training_landing_zone`
URL	ABFSS path to container	`abfss://container@account.dfs.core.windows.net/path`
Storage credential	Credential created in Part 3	`adls_training_credential`

> **Important:** Before creating the External Location, it is good practice to first create a dedicated container on ADLS for the target solution.

External Locations ›

Create a new external location

An external location allows you to access your data stored in cloud storage (e.g. Azure Data Lake Storage). You will need the cloud storage path and a paired credential (e.g. managed identity) which gives access to that path Learn more

External location name*

ext_altrn_storage

Storage type*

Azure Data Lake Storage

URL*

Enter the bucket path that you want to use as the external location. Note: This must be an ADLS Gen2 storage account with a hierarchical namespace

abfss://training@alttrnneadls.dfs.core.windows.net | Copy from DBFS

Storage credential* Learn more

Provide a storage credential capable of accessing the URL

ext_alttrn_storage (Managed Identity)

Connector Id: /subscriptions/8f8c2564-5635-4c2f-9b88-994145a10636/resourceGroups/mrg-ne-alt-training/pro...

User Assigned Managed Identity Id:

Create External Location — fill in the form

Click **Create** and wait for connection validation.

Create a new external location

An external location allows you to access your data stored in cloud storage (e.g. Azure Data Lake Storage). You will need the cloud storage path and a paired credential (e.g. managed identity) which gives access to that path Learn more

Location Type: Directory

- Success - **Read**
- Success - **List**
- Success - **Write**
- Success - **Delete**
- Success - **Path Exists**
- Success - **Hierarchical Namespace Enabled**

External Location — connection validated successfully ✓

Part 5: Configure Access Permissions

Final step — in the **Permissions** tab of the External Location, assign appropriate permissions to users/groups.

External Location — Permissions tab

Permission	Description
`READ FILES`	Read files from the location
`WRITE FILES`	Write files to the location
`CREATE EXTERNAL TABLE`	Create external tables pointing to this location
`ALL PRIVILEGES`	Full access — for dev/demo only!

> ⚠️ **Warning:** `ALL PRIVILEGES` was assigned for training purposes only.
> **Never use this in production!** Apply granular permissions aligned with your organisation's governance policy.

Summary — Configuration Flow

```
Azure Portal                          Databricks Workspace
____________                          ____________________

1. Managed Resource Group             3. Catalog → Credentials
   └─ Access Connector (Resource ID)     └─ Storage Credential
                                            (name + Resource ID)

2. Storage Account → IAM              4. Catalog → External Locations
   └─ Role Assignment                    └─ External Location
     (Managed Identity + RBAC Role)         (ABFSS URL + Credential)

                                      5. External Location → Permissions
                                         └─ Granular permissions per user/group
```

Concept	Description	Docs
Access Connector	Azure managed identity authenticating to storage on behalf of Databricks	docs.databricks.com/en/connect/storage/azure-storage.html

Storage Credential	UC object storing the cloud identity (Access Connector Resource ID)	docs.databricks.com/en/connect/unity-catalog/storage-credentials.html
External Location	UC object mapping an ABFSS path to a Storage Credential	docs.databricks.com/en/connect/unity-catalog/external-locations.html
ABFSS	Azure Blob FileSystem scheme: `abfss://container@account.dfs.core.windows.net/`	learn.microsoft.com/azure/storage/blobs/data-lake-storage-abfs-driver